

Regex rules

Match patterns in text

	example	matches
string	<i>hello</i>	hello
<code>\ </code> multiple string	<i>hello hi</i>	hello hi
<code>[]</code>	<i>b[aeiou]g</i>	big
<code>[-]</code> range	<i>a(2-4)z</i>	a2z
<code>.</code> any character	<i>a.z</i>	aQz
<code>^</code> start of line		
<code>\$</code> end of line		
repetition		
<code>*</code> 0+		
<code>+</code> 1+		
<code>+</code> ? 0 or 1		
<code>()</code> groups part of regex		
<code>\</code> escape	<i>example.com</i>	example.com

Commands

grep

- global regular expression print
- reads input line by line, if there's a match the entire line is printed with the match highlighted

`grep [options] regex [files]`

options:

<code>-r --recursive</code>	search in a directory and subdirectories
<code>-i --ignore-case</code>	non case sensitive

examples

```
ps -e | grep chrome
```

- see process number for processes called chrome
- -e shows all processes

- finds all files that contain string Brown
- searches recursively

```
james@laptop:~$ cat temp/file1.txt
555-2397 Brown, Barry
555-5116 Rogers, Sally
james@laptop:~$ grep -r Brown temp/*
temp/file1.txt:555-2397 Brown, Barry
```

```
grep -e "(something\.example\.com).*200" /etc/*
```

- -e extended regular expression rules, rather than just basic
- for some regex you need to use quotes, otherwise the shell can interpret it incorrectly
- Match the URL, then any number of characters before 200

sed

modifies a files contents

modifications are temporary unless you save them

```
sed [options] -f script-file [input-file]
```

```
sed [options] 'script-text' [input-file]
```

address

sed command operates on addresses, which are line numbers

- no address - operates on all lines
- one address
- range of addresses

options:

	address	
<code>a\text</code>	0 or 1	append text to file
<code>i\text</code>	0 or 1	text into file
<code>c\text</code>	range	replace selected lines with provided text
<code>s/regex/replacement</code>	range	replace the text that matches

example

```
sed 's/2012/2013/' calender1.txt calendar2.txt
```

- replace the first occurrence of 2012 on each line with 2013
- send the output in calendar2.txt